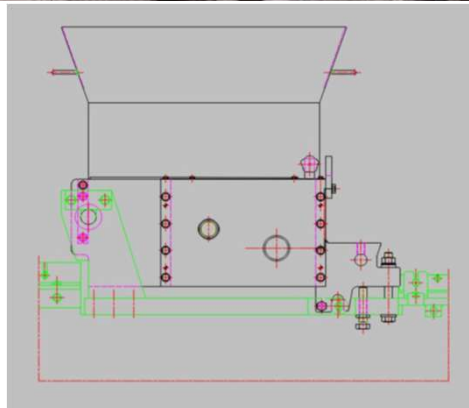


EXTRUSION HEAD + BENCH SECTION -TLE PROFILE-Relationship SOPs

MACHINE TYPE:TM101 SERIAL NUMBER :A.00146

FP McCANN (Cadeby)ORDER : C22004YEAR OF PRODUCTION 2023

- SOP A.00146.00 Contents List Extrusion Head TLE Profile
- SOP A.00146.0A Extrusion Head: Motor Drive System and Belt Tensionning
- SOP A.00146.0B Extrusion Head: Internal Side Guides
- SOP A.00146.0C Extrusion Head: Big and Small Slipper
- SOP A.00146.0D Extrusion Head: Big and Small Roller
- SOP A.00146.0E Extrusion Head: UCF Bearing Units and Drives
- SOP A.00146.0F Extrusion Head: Stirrer Assembly
- SOP A.00146.0G Extrusion Head: Hopper
- SOP A.00146.1 Extrusion Head: Removable Bench

EXTRUSION HEAD + BENCH SECTION -TLE PROFILE**MACHINE TYPE:TM101 SERIAL NUMBER :A.00146****EXTRUSION HEAD: MOTOR DRIVE SYSTEM AND BELT TENSIONING.****General Description and Function****MOTOR DRIVE SYSTEM AND
BELT TENSIONNING**

*This set of gearmotor and pulleys traction the Roller shaft of the various Extrusion Heads.

Critical Points and Risk Management

*It is essential to monitor the cleanliness of the gearmotor's traction pulley, located at the bottom of the extruder where concrete residue can accumulate.

*It is essential to monitor the as it is located at the bottom of the extruder where concrete residue can accumulate and put strain on the belt.

SUPERVISION LEVEL**DAILY**

289420 TOOTHED POOLEY (2 u)
289500 TOOTHED BELT

WEEKLY

289398 GEAR BOX + MOTOR
267096 BUSH
270131 BUSH
271186 BEARING (4 u)
289502 PNEUMATIC CYLINDER
289501 GUIDE UNIT

PREVENTIVE MAINTENANCE TASK**DAILY****CHECK:**

*Dirt buildup on the gearmotor and extruder traction pulleys.

*Wear condition of the belt.

*Air pressure for belt tension adjustment.

CLEANING**DAILY**

Clean with brushes and cloths:

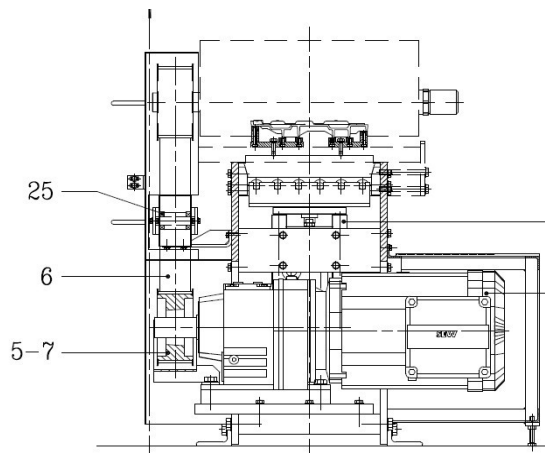
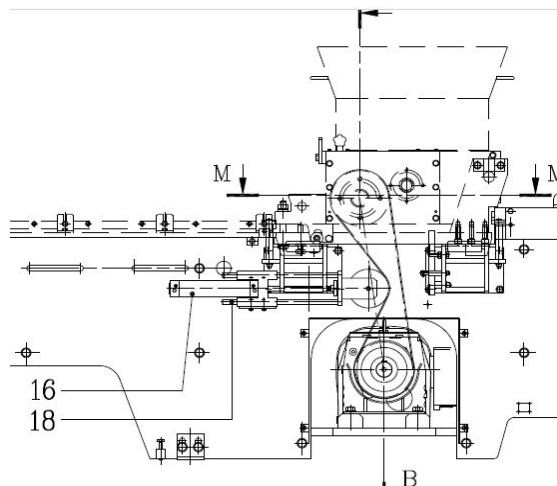
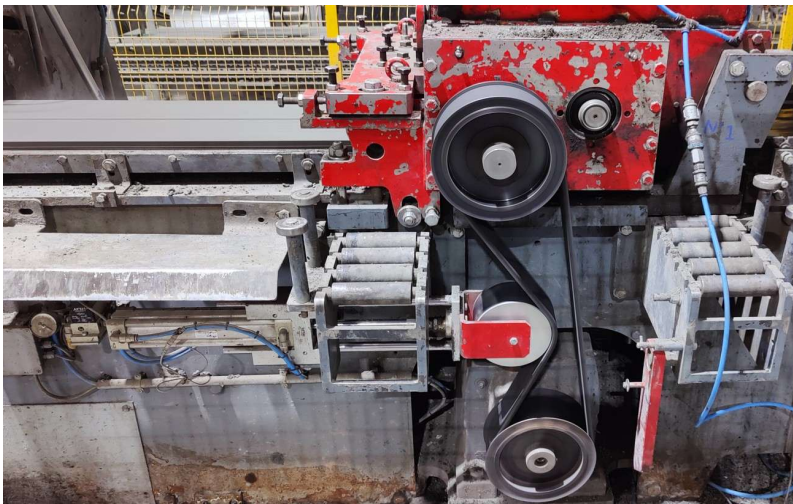
*The 2 traction pulleys

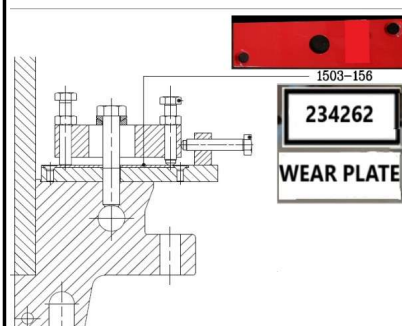
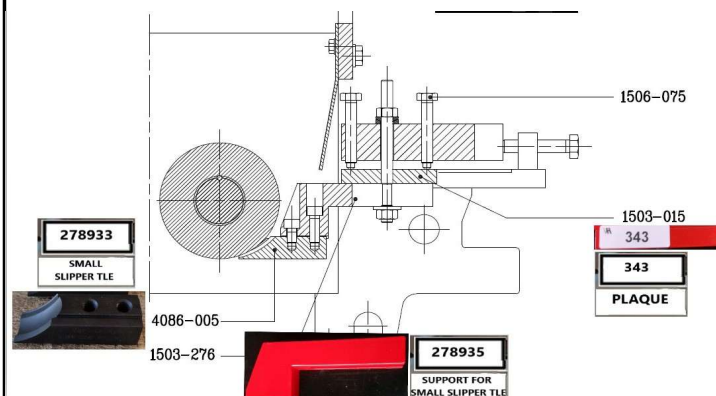
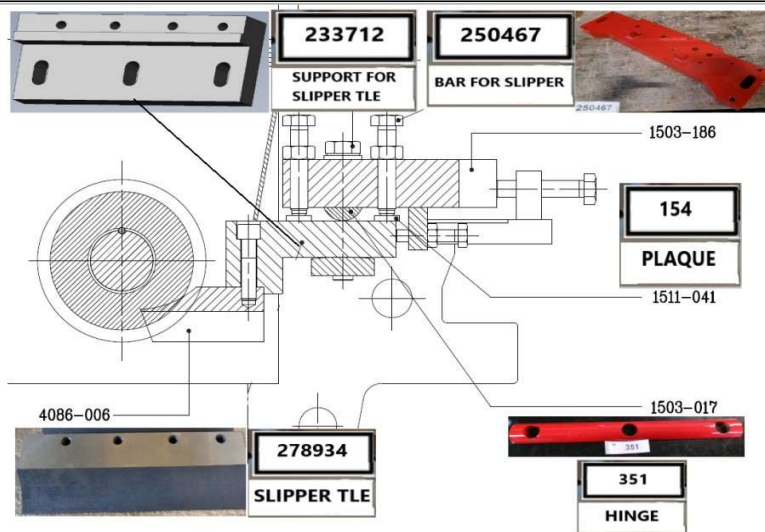
*The belt tensioning assembly Roller

WEEKLY

Clean with brushes and cloths:

*The cylinder and guides of the belt tensioning assembly.



EXTRUSION HEAD + BENCH SECTION -TLE PROFILE**MACHINE TYPE:TM101 SERIAL NUMBER :A.00146****EXTRUSION HEAD: BIG AND SMALL SLIPPER****General Description and Function****EXTRUSION HEAD SLIPPERS**

*These are the components responsible for the final shaping of the tile on the mould.

Critical Points and Risk Management

*The slippers are high-wear components ,identified by thickness reduction

*Regular monitoring and control or their thickness measurements are essential.

*Depending on whether the Slippers are made of steel or tungsten, the frequency of inspections will vary:
"DAILY or WEEKLY"

SUPERVISION LEVEL**DAILY**

278934 BIG SLIPPER TLE PROFILE
278933 SMALL SLIPPER TLE PRO

WEEKLY

250467 BAR FOR SLIPPERS
233712 SUPPORT SLIPPERS
154 PLAQUE 20 DIAMETER
351 HINGE
278935 SUPPORT SMALL SLIPPER
343 PLAQUE

PREVENTIVE MAINTENANCE TASK**DAILY****CHECK:**

*Inspect for cracks / deformations / /erosionon both Slippers' edges.

*Measure wear levels: Big & Small Slipper

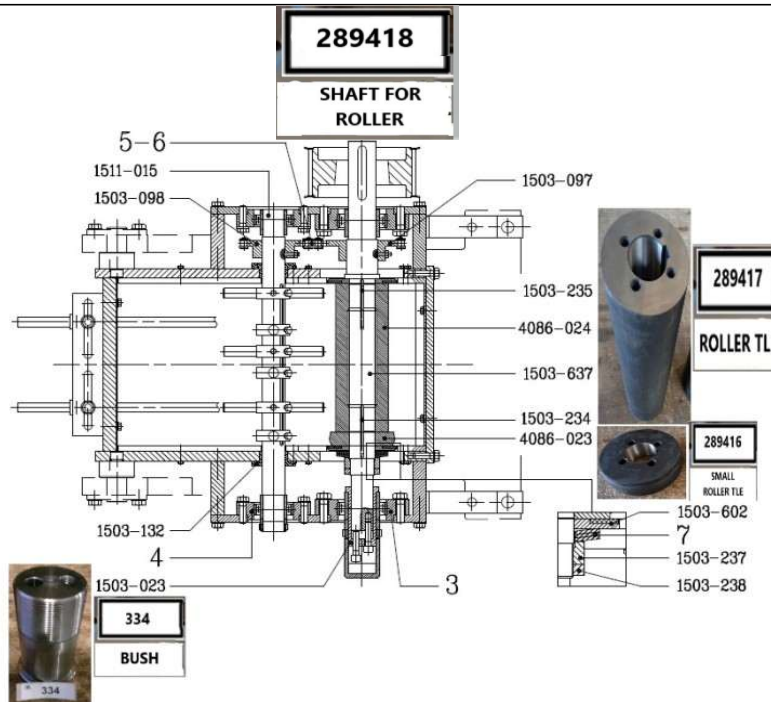
*Frequency:

Steel (DAILY) / Tungsten (WEEKLY)

CLEANING**DAILY**

*Remove debris from all guide surface bar, supports , slippers using scrapers and clean cloths.

* Finish by brushing on mould release oil.

EXTRUSION HEAD + BENCH SECTION -TLE PROFILE**MACHINE TYPE:TM101 SERIAL NUMBER :A.00146****EXTRUSION HEAD: BIG AND SMALL ROLLER****General Description and Function****EXTRUSION HEAD ROLLERS**

*These initial elements perform the tile pre-forming stage prior to final shaping in the Slippers assembly.

Critical Points and Risk Management

*The Big Roller, Small Roller, and both side Discs are high-wear components, identifiable by diameter reduction.

*Regular monitoring and control of diameter measurements at multiple points on both rollers is essential.

*Depending on whether the Rollers are made of steel or tungsten, the frequency of inspections will vary:
"DAILY or WEEKLY"

SUPERVISION LEVEL**DAILY**

289416 SMALL ROLLER TLE PROF
289417 ROLLER TLE PROFILE
281504 LATERAL DISC (2 u)

WEEKLY

289418 SHAFT FOR ROLLER
334 BUSH

PREVENTIVE MAINTENANCE TASK**DAILY****CHECK:**

*Inspect for cracks/deformations/erosion on both Rollers' edges.

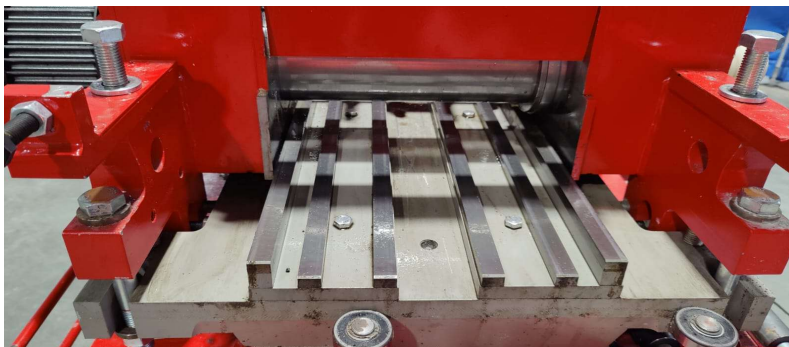
*Measure wear levels: Big & Small Rollers

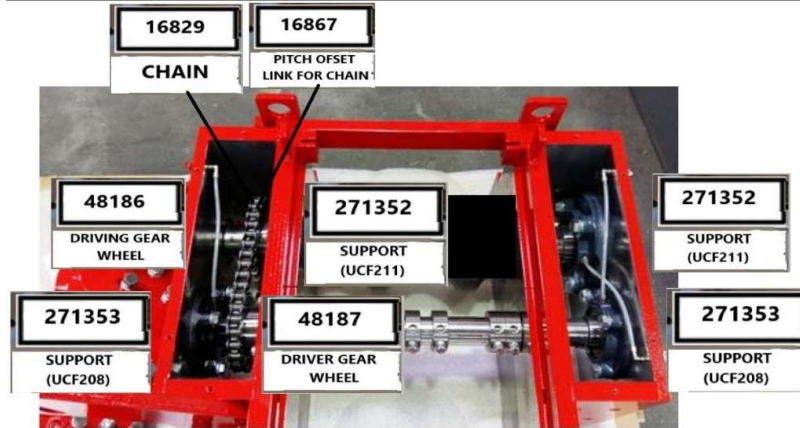
*Frequency:
Steel (DAILY) / Tungsten (WEEKLY)

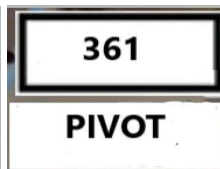
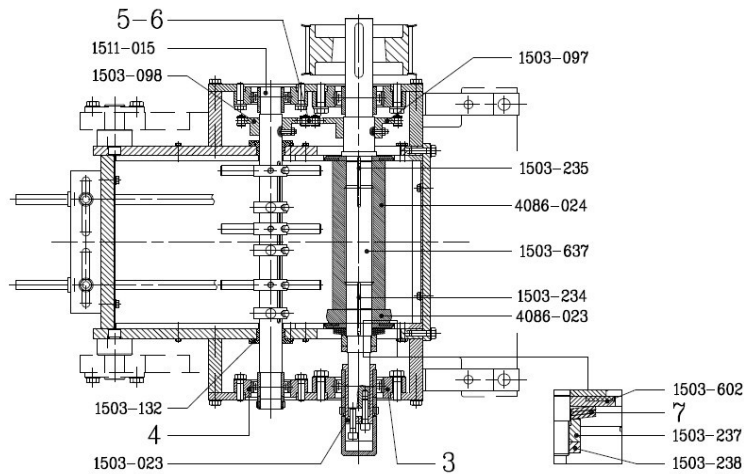
CLEANING**DAILY**

*Remove debris Rollers and Lateral Discs using scrapers and clean cloths.

* Finish by brushing on mold release oil.



VORTEX HYDRA	STANDARD OPERATING PROCEDURE	Code: SOP A.00146.0E Date: 15/12/25 Page: 6 de 9
EXTRUSION HEAD + BENCH SECTION -TLE PROFILE		
MACHINE TYPE:TM101 SERIAL NUMBER :A.00146		
EXTRUSION HEAD: UCF BEARING UNITS AND DRIVES	<u>General Description and Function</u>	
 ROLLER LATERAL FLANGE	<u>UCF BEARING UNITS AND DRIVES</u>	
 16829 CHAIN 16867 PITCH OFFSET LINK FOR CHAIN 48186 DRIVING GEAR WHEEL 271352 SUPPORT (UCF211) 271352 SUPPORT (UCF211) 271353 SUPPORT (UCF208) 48187 DRIVER GEAR WHEEL 271353 SUPPORT (UCF208)	*These are the components that provide support and traction to the Roller and Stirrer shafts.	
		<u>Critical Points and Risk Management</u>
		*UCF bearing units must be monitored, especially the Roller's unit as it significantly higher stress compared to the Stirrer's.
		*The pinion driving the Stirrer shaft from the Roller shaft requires monitoring , including chain tension and condition checks.
		*The 2 covers housing the 4 UCF bearing units and pinion gears must remain closed to prevent external contamination.
		DAILY 48186 DRIVING GEAR WHEEL 48187 DRIVEN GEAR WHEEL 16829 CHAIN 16867 PITCH 271352 SUPPORT (UCF211) 271353 SUPPORT (UCF208)
		<u>PREVENTIVE MAINTENANCE TASK</u>
		DAILY
		<u>CHECK:</u>
		*Wear level of pinion gears, chain, and spline coupling
		*Lubricate all UCF bearing units.
		*Evaluate required frequency
		*Adjust lubrication intervals based on machine operational demands.
		*No grease should leak externally during lubrication of all UCF bearing units.
		CLEANING
		WEEKLY
		*Clean all externally accessible areas of the pinion gears and UCF bearing units brushes and rags.

EXTRUSION HEAD + BENCH SECTION -TLE PROFILE**MACHINE TYPE:TM101 SERIAL NUMBER :A.00146****EXTRUSION HEAD: STIRRER ASSEMBLY****General Description and Function****EXTRUSION HEAD STIRRER
ASSEMBLY**

*The Stirrer shaft with its pivots feeds concrete to the Rollers and Slippers for proper tile formation- both are high-wear components.

Critical Points and Risk Management

*The pivots are initially made of round steel.

Based on feed requirements, their capacity can be increased by:

*For high-wear applications, they can be tungsten-coated.

SUPERVISION LEVEL**DAILY**

361 PIVOT (12 u)

174 COLLAR (6 u)

165 SHAFT FOR STIRRER

WEEKLY

58800 WEAR COLLAR (2 u)

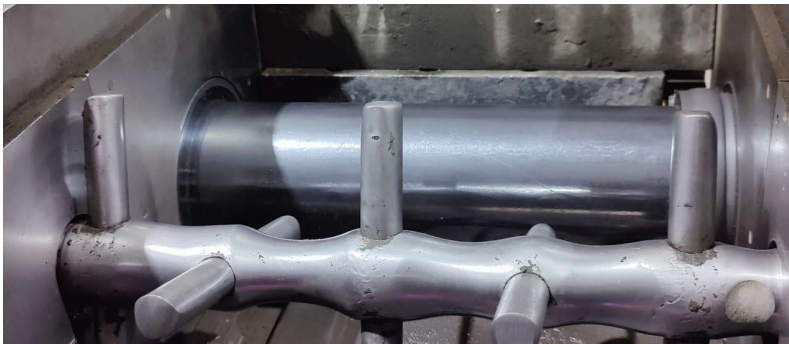
PREVENTIVE MAINTENANCE TASK**DAILY****CHECK:**

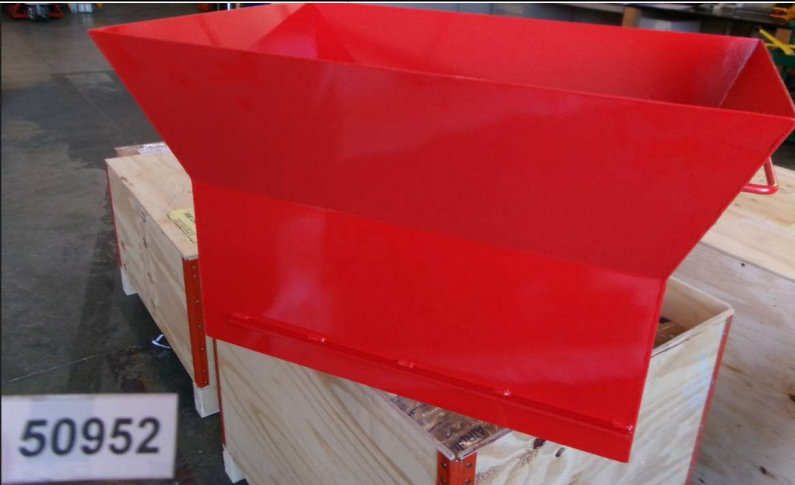
*Wear and erosion levels of the Shaft, Collars, and Pivots.

CLEANING**DAILY**

*Clean all shaft, collar, and pivot areas using scrapers and rags.

* Finish by brushing on mold release oil.



VORTEX HYDRA	STANDARD OPERATING PROCEDURE	Code: SOP A.00146.0G Date: 15/12/25 Page: 8 de 9
EXTRUSION HEAD + BENCH SECTION -TLE PROFILE		
MACHINE TYPE:TM101 SERIAL NUMBER :A.00146		
EXTRUSION HEAD: HOPPER		<u>General Description and Function</u>
		<u>EXTRUSION HEAD HOPPER</u> *This is the concrete storage zone that feeds material to the Rollers and Slippers via the Stirrers.
		<u>Critical Points and Risk Management</u> *Maintaining a constant concrete level in the hopper is critical for consistent material feed during tile formation. *If excessive concrete adheres to the hopper surfaces, consider lining the hopper with a low-adhesion material such as Duroglis.
		<u>SUPERVISION LEVEL</u>
		50592 HOPPER
		<u>PREVENTIVE MAINTENANCE TASK</u>
		<u>DAILY</u>
		<u>CHECK:</u>
		*The surfaces of the hopper.
		<u>CLEANING</u>
		<u>DAILY</u>
		*Clean the Hopper using scrapers.
		* Finish by brushing on mold release oil.

EXTRUSION HEAD + BENCH SECTION -TLE PROFILE -Relationship SOP**MACHINE TYPE:TM101 SERIAL NUMBER :A.00146****EXTRUSION HEAD: REMOVABLE BENCH****General Description and Function****REMOVABLE BENCH**

This removable bench assembly consists of 12 guides:

***The first 6 guides are steel**

***The remaining 6 guides are tungsten.**

***The first 6 guides facilitate the transition of the Push Chain to the Extruder Box, while the remaining 6 guides bear the pressure from the Roller and Slipper during tile formation in the extrusion process.**

Critical Points and Risk Management

***The 6 tungsten guides require rigorous supervision - any fracture or erosion can cause irreparable damage to mould base.**

SUPERVISION LEVEL**DAILY**

28898 PALLET GUIDE 20x12x600 (6 u)

434 PALLET GUIDE 20x12x200 (6 u)

2085 SHIM H=33,5 (1 u)

2070 SHIM H=19 (1 u)

PREVENTIVE MAINTENANCE TASK**DAILY****CHECK:**

***Surfaces & Contours of the 6 Pallet Guides 28898**

CLEANING**WEEKLY**

***Clean with brushes and cloths on the surfaces of all 12 guides.**

